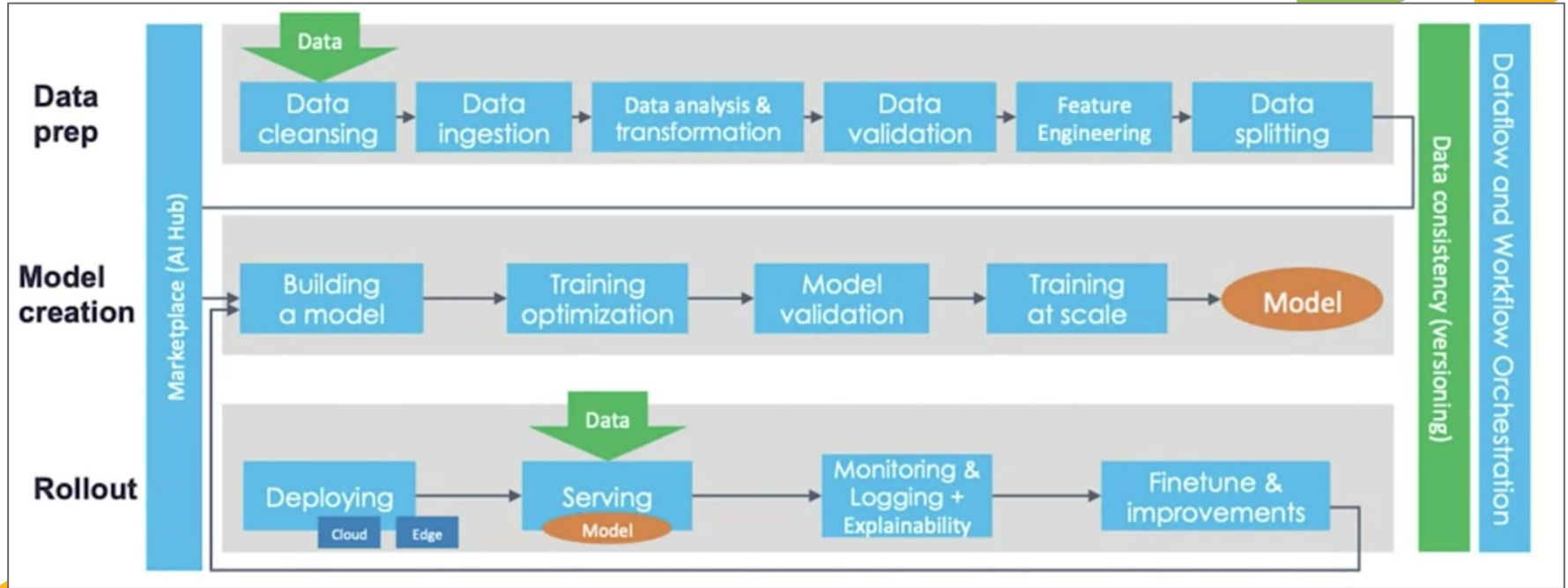


Data Lifecycle

by LAB Team

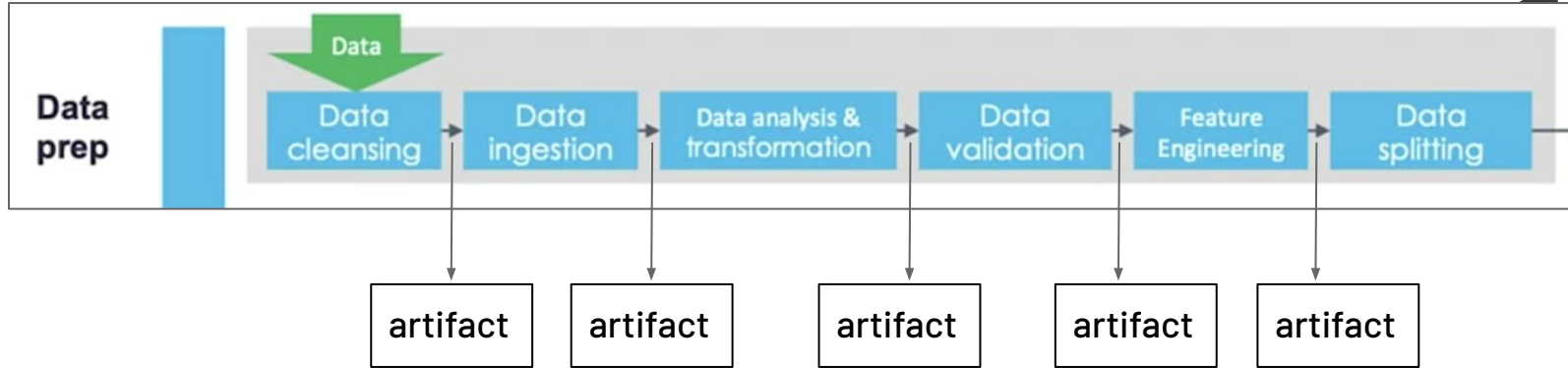


Overall



The data journey

Interpreting model results requires understanding data transformation



Artifacts include all of the objects (data in different transformation, models, metrics, etc.) produced by pipeline components

Data provenance and lineage

Data provenance and lineage is a sequence of the artifacts that are created through the pipeline



Inspect artifacts at each point in the training process



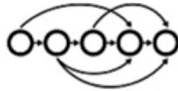
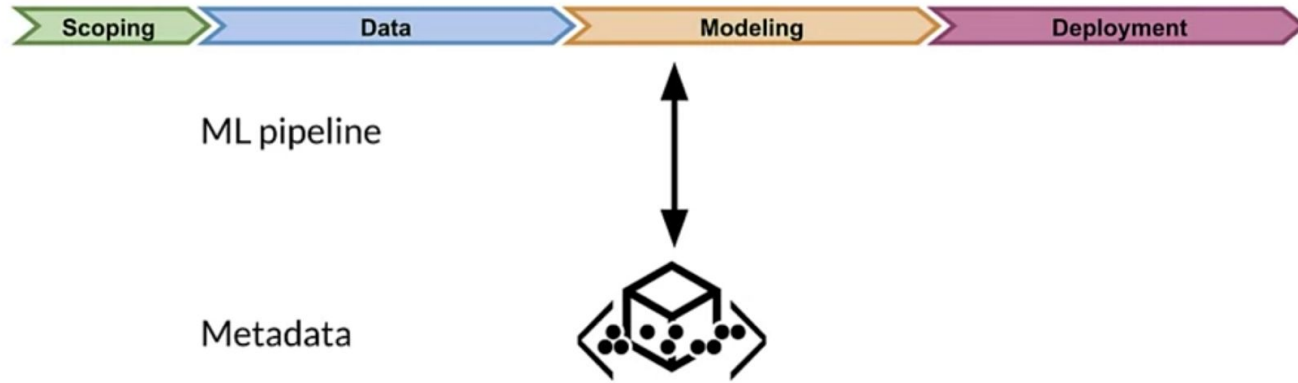
Trace back through a training run



Compare training runs

➡ Data versioning is data files version control to track data provenance and lineage

ML Metadata



Produce DAG of pipelines



Verify the inputs used in an execution



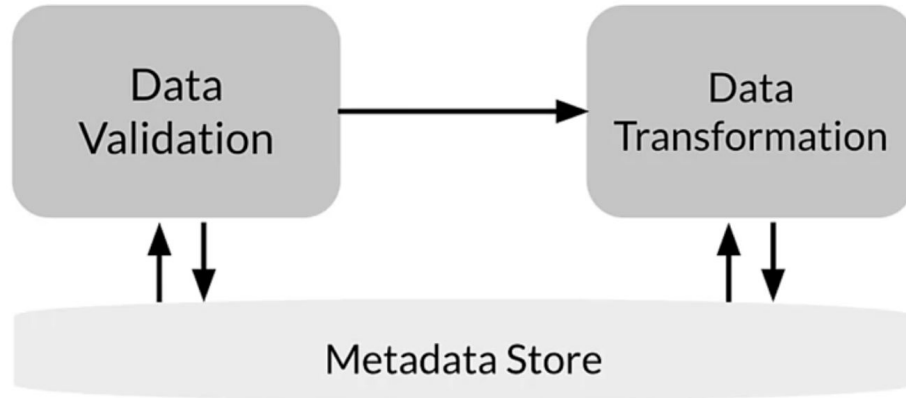
List all artifacts



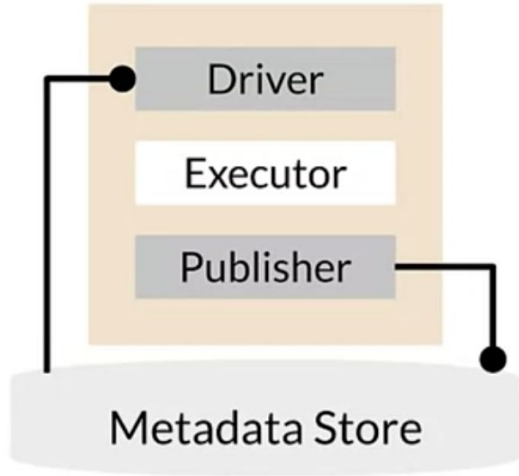
Compare artifacts

ML Metadata

Example: Everything we use as an input to “Data validation” function and “Data Transformation” function and their output results will be store at metadata store



ML Metadata



- Driver:
 - Supplies required metadata to executor
- Executor:
 - Place to code the functionality of component
- Publisher:
 - Stores result into metadata

ML Metadata



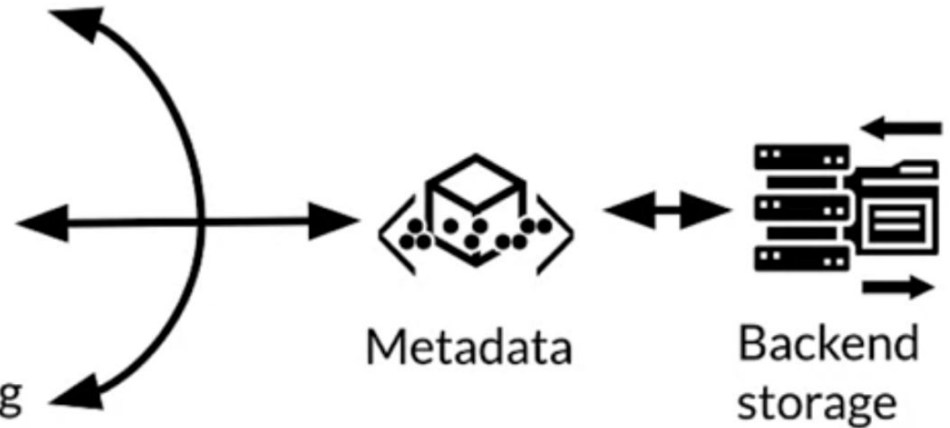
Artifacts: Data going as input or generated as output by a component



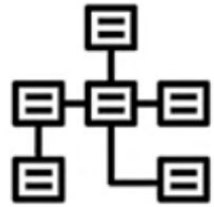
Execution: Record of component in pipeline.



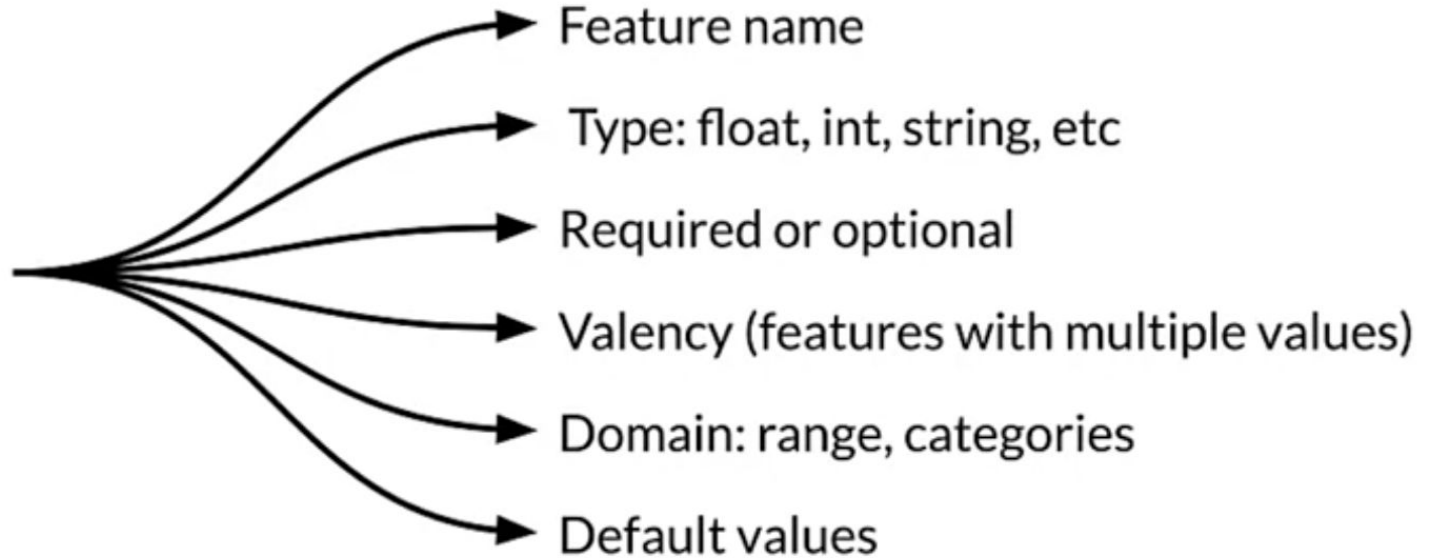
Context: Conceptual grouping of executions and artifacts.



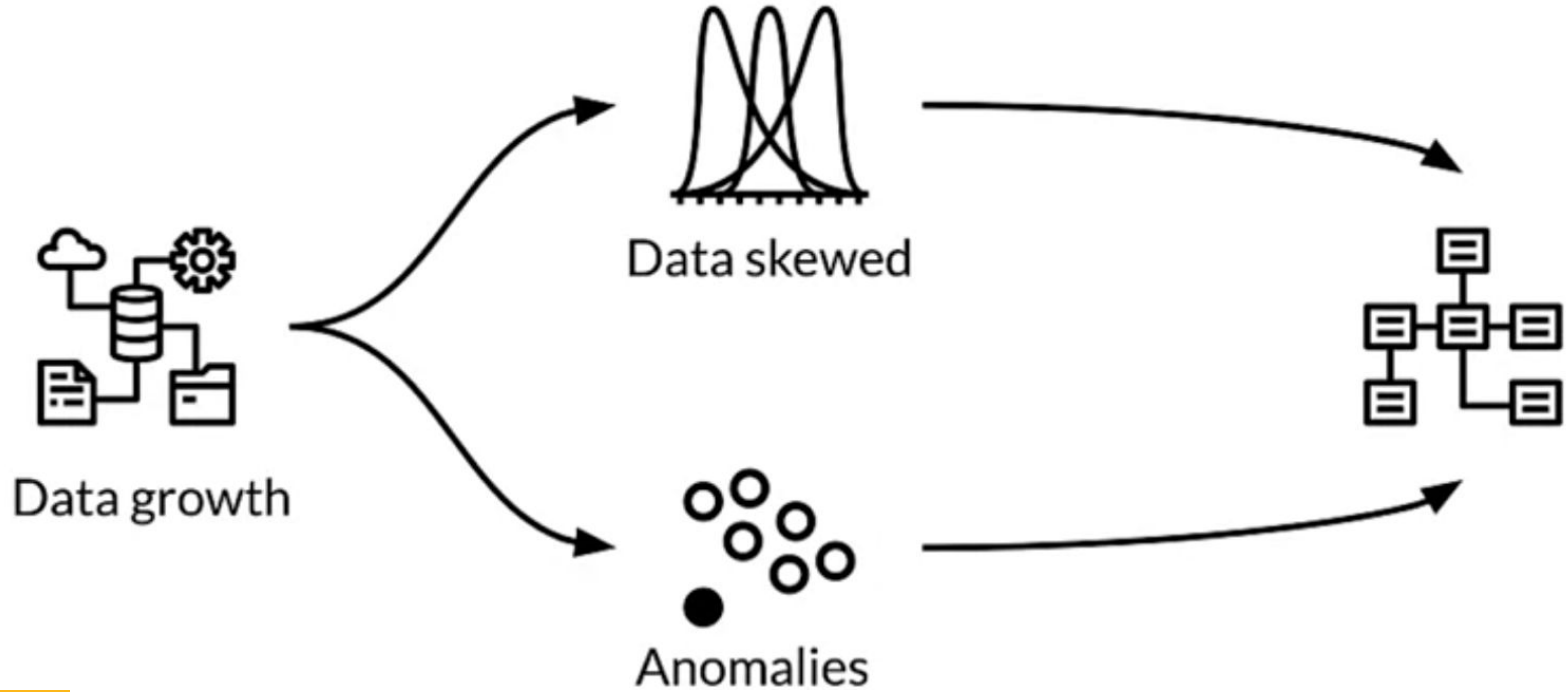
Schema



Schema



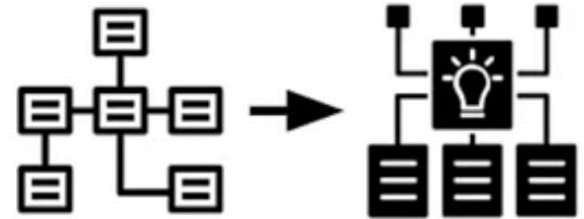
Schema



Schema



Looking at schema versions to
track data evolution



Schema can drive other
automated processes

